

ERIC SUN

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EDUCATION

University of Pennsylvania

M.S. Physics, *Submatriculant (Joint B.A./M.S.)*
B.A. Physics (Hons.), *Concentration in Computational Techniques*
B.S.E. Computer Science
Minor in Mathematics

August 2023 – May 2027

Philadelphia, Pennsylvania

GPA: 3.94 / 4.00, Vagelos Integrated Program in Energy Research (VIPER)

Select Coursework: Machine Learning (G), Spatial Network Theory (G), Graduate Quantum Mechanics I / II (G), Big Data Processing (G), Natural Language Processing (UG), Operating Systems (UG), Complex Analysis (UG), Statistical Mechanics (UG)

EXPERIENCE

Citadel LLC

Quantitative Development Intern, Physical Natural Gas Desk

June 2026 – August 2026

New York Metropolitan Area

- Architected and built a new multi-threaded valuation service with Python and C++ bindings, Celery task orchestration, Redis, and Kubernetes for Citadel Energy Marketing, speeding up risk sensitivities (present value, delta, gamma, vega, etc.) of exotic natural gas products (transport contracts, spark spreads, etc.) by 53x compared to the prior framework.
- Expanded the natural gas deal booking platform to support structured products and visualize pre-booking risk using Plotly and Ag-Grid, now the default platform for natural gas off-exchange booking, servicing over 2000 deals daily.
- Created an agentic AI assistant with Gemini API and LangChain that can value, analyze, and book new deals automatically, reducing post-trade deal-entry bottleneck from 1-2 minutes of manual entry to 3-5 seconds.

University of Pennsylvania

Student Researcher, Andrew M. Rappe Group

January 2024 – January 2026

Philadelphia, Pennsylvania

- Led computational quantum mechanics project to use physics-informed neural networks (PINNs) on PyTorch to optimize norm-conserving fully nonlocal pseudopotentials with relaxed boundary criteria.
- Wrote a multi-threaded C++ library that solves the Kohn-Sham equations using a self-consistent field in less than 250 ms and 10^{-9} Hartree tolerance up to Uranium, supporting hybrid functionals and relativistic corrections with LibXC.
- Refactored 50k lines of self-consistent field density functional theory (DFT) routines for the OPIUM library in C and FORTRAN, including writing over 15k lines of new Kohn-Sham, Hartree-Fock, and Dirac-Fock solvers.

Amazon Web Services (AWS)

Software Development Engineer (SDE) Intern, Center for Quantum Computing

May 2025 – August 2025

San Francisco, California

- Created machine learning pipelines that deployed agentic models for creating, executing, and processing experiments on quantum computers using AWS Bedrock, Prefect, PostgreSQL, SQLAlchemy, and GraphQL, used by over 250 scientists at the AWS Center for Quantum Computing and the Oscar Painter Group at Caltech.
- Implemented retrieval augmented generation (RAG) via the Model Context Protocol (MCP) with FastMCP, using abstract syntax tree (AST) traversal to boost zero-shot accuracy for scientific code generation from 15% to 95%.
- Used Docker and Selenium to deploy automated web scrapers running on AWS Lambda which can populate RAG databases built with AWS OpenSearch and AWS S3.

Yale University

Student Researcher & Summer Session Instructor, BCT 326 Group

June 2022 – January 2023

New Haven, Connecticut

- Wrote clustering tools for angle-resolved photoemission spectroscopy (ARPES) data of correlated insulators and cuprate superconductors with convolutional autoencoders built using TensorFlow and SciPy, improving clustering results over standard principal component analysis (PCA) by 8.86%
- Generated artificial training data by applying Poisson sampling to tight-binding spectral bands of topological materials.
- Co-instructed condensed matter enrichment course for the 2022 Yale Pathways to Science summer program.

SELECT AWARDS

United States Physics Olympiad (USAPhO) Semifinalist, Top 200 (2023)
Tournament of Champions National Finalist, Parliamentary Debate (2023)
AAPT PhysicsBowl World Ranked 98th/5000+, Captain of Highest Ranked US Team (2022)

TECHNICAL SKILLS

Languages: Python, C, C++, FORTRAN, Bash, Zsh, SQL, $\LaTeX$

Libraries: PyTorch, Scikit-learn, Pandas, Polars, NetworkX, OpenCV, NumPy, SciPy, CUDA, TensorFlow, Keras, Sphinx, Matplotlib, Selenium, FastAPI, FastMCP, SQLAlchemy, Prefect, MyPy, Ruff, Pydantic, ChromaDB, Pybind11, Numba

Tools: Git, Jupyter, CI/CD, Docker, PostgreSQL, MySQL, Slurm, AWS (Bedrock, Lambda, EC2, S3, Aurora, Neptune)